CURRICULUM VITAE

PUNEET PAREEK

Laboratoire Interdisciplinaire de Physique (LIPhy), Université Grenoble Alpes, France
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RESEARCH INTERESTS

• Glassy Dynamics • Disordered Systems • Active Matter • Non-equilibrium Statistical Mechanics
• Biophysics

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher Current — Oct 2026 – Present
Institute: Laboratoire Interdisciplinaire de Physique (LIPhy), Université Grenoble Alpes, Grenoble, France

Postdoctoral Fellow — Apr 2026 – Sep 2026
Institute: Tata Institute of Fundamental Research (TIFR), Mumbai, India
Supervisor: Prof. Saroj Kumar Nandi, TIFR Hyderabad, India

Research Associate — Oct 2025 – Apr 2026
Institute: Tata Institute of Fundamental Research (TIFR), Mumbai, India
Supervisor: Prof. Saroj Kumar Nandi, TIFR Hyderabad, India

EDUCATION

Ph.D. Physics — 2020 – 2026
University: Tata Institute of Fundamental Research (TIFR), Mumbai, India
Thesis Title: Characteristics of glassy dynamics and how active self-propelled forces affect them
Supervisor: Prof. Saroj Kumar Nandi, TIFR Hyderabad, India

M.Sc. Physics — 2018 – 2020
University: Indian Institute of Technology Delhi (IIT-D), New Delhi, India
Project Title: Shortcuts to adiabaticity with active particles
Supervisor: Prof. Rahul Suresh Marathe, IIT-D, India
CGPA: 9.507 / 10.000

B.Sc. (H) Physics — 2015 – 2018
University: University of Delhi, Delhi, India
CGPA: 9.203 / 10.000

STATEMENT OF PAST RESEARCH

- **Various time scales in passive glassy dynamics:** We found that caging controls the mean, whereas dynamical heterogeneity (DH) governs the distribution of particle displacements. In addition, we can define several other time scales, such as when the mean-square displacement becomes diffusive, the distribution of particle displacement becomes Gaussian, the overlap function becomes a power-law, the intermediate scattering function becomes exponential, etc. We demonstrated using analytical calculations and simulation results that all these different timescales are related to either caging or DH. This naturally explains the Fickian yet non-Gaussian behaviour of glasses.

- **Crossover in autocorrelation function from stretched exponential to power-law:** The overlap function in passive glasses shows a crossover from stretched exponential (SE) at short times to a power law at long times. To understand the mechanism behind such a crossover, we defined a simplified one-dimensional spin model that belongs to the class of kinetically constrained models. The model has an onsite energy function and a kinetic rule that a spin can flip only if the neighbouring spins have opposite signs. Thus, the dynamics conserve energy, which comes from the movements of domain walls (DWs). We found that the crossover comes from a change in the relevant excitations: from DWs at short times to bound pairs of DWs at long times.
- **Influence of activity on relaxation dynamics and fragility in active glasses:** Active glasses comprise self-propelled particles with a self-propulsion force f_0 and persistence time τ_p and provide a minimal model for biological systems. They often show nontrivial behaviors, such as reentrant dynamics and tunable fragility. We studied a dense active system of soft repulsive particles that shows these features. We characterised the critical surface separating liquid and glassy states and showed that the nontrivial features are consequences of the morphology of this surface. In addition, with increasing τ_p, the nature of the critical point changes from glass-like to jamming-like. The tunable fragility, where the dynamics goes from sub-Arrhenius to super-Arrhenius, also appears in biological tissues. We employed a dynamic scaling analysis and showed that this behavior in the current model is governed by an “active glass line”.

LIST OF PUBLICATIONS

Published

- **Puneet Pareek**, Peter Sollich, Saroj Kumar Nandi, Ludovic Berthier, Scaling the glassy dynamics of active particles: Tunable fragility and reentrance, *Proc. Natl. Acad. Sci. U.S.A.* **123** (4) e2516624123 (2025).
- Sukanta Mukherjee, **Puneet Pareek**, Mustansir Barma, Saroj Kumar Nandi, Stretched exponential to power-law: crossover of relaxation in a kinetically constrained model, *J. Stat. Mech.* (2024) 023205.
- **Puneet Pareek**, Monoj Adhikari, Chandan Dasgupta, Saroj Kumar Nandi, Different glassy characteristics are related to either caging or dynamical heterogeneity, *J. Chem. Phys.* **159**, 174503 (2023).

Under Review

- Soumitra Kolya, **Puneet Pareek**, Saroj Kumar Nandi, Active Inhomogeneous Mode-Coupling Theory (aIMCT) for Dense Systems of Self-Propelled Particles, arXiv:2410.15928 (2024).

Under Preparation

- **Puneet Pareek**, Saroj Kumar Nandi, Non-trivial effect of fraction of active particles on glassy dynamics.

Thesis

- **Puneet Pareek**, Characteristics of glassy dynamics and how active self-propelled forces affect them, Ph.D. Thesis, Tata Institute of Fundamental Research (2026).

CONFERENCES ATTENDED

- **The Discussion Meeting on Glassy, Disordered and Non-equilibrium Soft Matter**, 10–11 April 2025
  Raman Research Institute, Bangalore, India
  Poster title: *Scaling the glassy dynamics of active particles: Tunable fragility and reentrance.*
- **Physics of Life: Active and Living Matter**, 8–10 February 2024
  Institute of Physics, Bhubaneswar, India
  Poster title: *Re-entrance of Relaxation Dynamics in Dense Assemblies of Active Particles.*
- **Frontiers in Active and Soft Matter 2023**, 10–11 February 2023
  Tata Institute of Fundamental Research Hyderabad, India
  Poster title: *Can Glasses be Fickian yet non-Gaussian?* (Best Poster Award)

- **In-house Symposium**, 15–17 October 2022
  Tata Institute of Fundamental Research Hyderabad, India
  Poster title: *Stretched Exponential to Power Law in Glassy Systems.*

AWARDS AND SCHOLARSHIPS

- 2023: Best Poster Award in the conference “Frontiers in Active and Soft Matter” (2023).
- 2019–20: Merit-cum-Means and Prof. Vidya Bhushan Scholarship at Indian Institute of Technology Delhi.
- 2017–18: Science Meritorious Award at University of Delhi.

ACHIEVEMENTS

- 2020: Graduate Aptitude Test in Engineering (GATE) Physics with All India Rank 7
- 2020: Joint Entrance Screening Test (JEST) with All India Rank 59
- 2020: CSIR Junior Research Fellowship (JRF) Physics with All India Rank 76
- 2018: Joint Admission Test for M.Sc. (JAM) Physics with All India Rank 106

SOFTWARE AND COMPUTATIONAL SKILLS

Programming:	Python, C, C++
Software:	LAMMPS, OVITO, Mathematica, LaTeX, GNUplot
Operating systems:	Linux, Microsoft Windows

REFEREES

Dr. Saroj Kumar Nandi
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Email: saroj@tifrh.res.in

Prof. Chandan Dasgupta
Honorary Professor, Indian Institute of Science, Bangalore, India
ICTS Endowed Visiting Professor, International Centre for Theoretical Sciences, Bangalore, India
Email: cdgupta@iisc.ac.in

Prof. Mustansir Barma
Professor Emeritus
Tata Institute of Fundamental Research Hyderabad, India
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